Example 2.11 Give any two rational numbers lying between 0.5151151115…. and 0.5353353335…

Solution Two rational numbers between the given two irrational numbers are 0.5152 and 0.5352

Example 2.12 Find whether x and y are rational or irrational in the following.

(i) $a = 2 + \sqrt{3}$, $b = 2 - \sqrt{3}$; $x = a + b$, $y = a - b$

(ii) $a = \sqrt{2} + 7$, $b = \sqrt{2} - 7$; $x = a + b$, $y = a - b$

(iii) $a = \sqrt{75}$, $b = \sqrt{3}$; $x = ab$, $y = \frac{a}{b}$

(iv) $a = \sqrt{18}$, $b = \sqrt{3}$; $x = ab$, $y = \frac{a}{b}$

Note

From these examples, it is clear that the sum , difference, product, quotient of any two irrational numbers could be rational or irrational.

Solution

(i) Given that $a = 2 + \sqrt{3}, b = 2 - \sqrt{3}$

$x = a + b = (2 + \sqrt{3}) + (2 - \sqrt{3}) = 4$, a rational number.

$y = a - b = (2 + \sqrt{3}) - (2 - \sqrt{3}) = 2\sqrt{3}$, an irrational number.

(ii) Given that $a = \sqrt{2} + 7$, $b = \sqrt{2} - 7$

$x = a + b = (\sqrt{2} + 7) + (\sqrt{2} - 7) = 2\sqrt{2}$, an irrational number.

$y = a - b = (\sqrt{2} + 7) - (\sqrt{2} - 7) = 14$, a rational number.

(iii) Given that $a = \sqrt{75}$, $b = \sqrt{3}$

$x = ab = \sqrt{75} \times \sqrt{3} = \sqrt{75 \times 3} = \sqrt{5 \times 5 \times 3 \times 3} = 5 \times 3 = 15$, a rational number.

$y = \frac{a}{b} = \frac{\sqrt{75}}{\sqrt{3}} = \sqrt{\frac{75}{3}} = \sqrt{25} = 5$, rational number.

(iv) Given that $a = \sqrt{18}$, $b = \sqrt{3}$

$x = ab = \sqrt{18} \times \sqrt{3} = \sqrt{18 \times 3} = \sqrt{6 \times 3 \times 3} = 3\sqrt{6}$, an irrational number.

$y = \frac{a}{b} = \frac{\sqrt{18}}{\sqrt{3}} = \sqrt{\frac{18}{3}} = \sqrt{6}$, an irrational number.

Example 2.13 Represent $\sqrt{9.3}$ on a number line.

Solution

- Draw a line and mark a point A on it.
- Mark a point B such that $AB = 9.3$ cm.
- Mark a point C on this line such that $BC = 1$ cm.
- Find the midpoint of AC by drawing perpendicular bisector of AC and let it be O
- With O as center and $OC = OA$ as radius, draw a semicircle.

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